

Education

B.S. Mechanical Engineering, Tau Beta Pi, University of California, Berkeley

Exp Graduation: May 2028 | GPA: 3.91

Relevant Experience

Component Test Engineering Intern (Incoming) • Tesla

Fall 2026

- Accepted offer to join Component Engineering team, supporting automated camera & optics test systems
- Designed modular test chamber system for monitoring and ventilation gas LEL levels to adhere to EH&S safety standards

Manufacturing & Automation Engineering Intern • 10x Genomics

Summer 2026

Clamshell Pinning Press

- Designed pneumatic mold-pinning press, cutting operator pinning time 62% (12s → 4.5s), saving ~160 operator-hours/yr
- Built and tuned a two-hand safety relay/timer circuit with pneumatic pressure and flow controls, achieving 100% pin-placement success with safe, controlled actuation and drive-force margin.
- Authored GD&T drawings specifying dowel-pin slip- and press-fit tolerances to eliminate fixture wobble, driving fabrication of 12+ custom 6061-T6 components.

50L Carboy Washing Fixture

- Designed rotating carboy-washing fixture with adjustable magnetic torque limiter (2–50 in-lb), eliminating an EH&S-flagged manual-lifting hazard and cutting wash time 50% (4 min → 2 min), reclaiming ~15–25 operator-hours annually.
- Prototyped a Schatz-linkage figure-8 mechanism, then simplified to a final design using 1/3 the parts and fewer moving points of failure, cutting complexity while meeting the same coating-coverage requirement.

Palomar Die Bonder Dispensing Improvement

- Diagnosed die-placement alignment failures to excessive breakaway friction in the existing pneumatic actuator and specced a low-friction air cylinder for precise, low-pressure standstill positioning; the spring-return configuration halved solenoid count versus the prior double-acting design.
- Restructured the PLC control logic to drive the new single-acting (spring-return) actuation cycle in place of double-acting control, integrating the updated sequence into the die-bonder's existing robot/PLC handshake.

Manufacturing Lead, Chassis Engineer • Formula Electric at Berkeley

Sept 2024 - Present

- Led design of 2025 steel spaceframe, running parametric ANSYS sweeps on tube sizing/geometry to hit 1400 Nm/deg torsional stiffness target while cutting cross-sectional area 7.1% and boosting MOI 6.9% vs. prior chassis
- Designed torsional stiffness test rig, validating FEA within 11.9% of physical measurement
- Built Python vehicle dynamics sensitivity models linking chassis torsional stiffness to suspension compliance and LLTD, guiding handling/tunability decisions
- Managed fabrication as Lead Welder, reducing total weld cycle time from 6 to 4 weeks through streamlined fixturing and utilizing GD&T to ensure precision suspension tab alignment.
- Conducted ASTM-standard 3-point bend and dye penetrant tests for rule-compliant welded/composite samples
- Hand-laid wet-layup carbon fiber wings and structural panels, vacuum bagging parts and developing test procedure to tune resin content and vacuum settings for consistent laminate quality

Lead Research Engineer • Theoretical & Applied Fluid Dynamics Laboratory (TAFLAB)

May 2025 - Present

- Conducted multiphysics optimization in ANSYS Maxwell for a low-frequency (0.3 Hz) electromagnetic harvester; achieved 200mW avg power, satisfying 20% of the total power requirement for autonomous surface vehicle (ASV) sensor suites.
- Prototyped and optimized oscillating electromagnetic energy harvester in ANSYS Maxwell to <6.3% validated error
- Modeled oscillatory system dynamics in MATLAB with Lagrangian mechanics to select spring stiffness for resonance tuning
- Created custom coil winding jig and rectification circuitry to convert AC output into regulated DC for battery charging
- Engineered redundant, multi-layer waterproofing to protect onboard PCBs from ocean environment ingress, cutting ASV failure rate from 1-in-5 to 1-in-10 units.

Propulsion Manufacturing Engineer • UC Berkeley's Space Technology and Rocketry (STAR)

Sept 2024 - May 2025

- L2 Rocket: Designed and fabricated with integrated avionics, achieving successful launch and recovery at 2,944 ft

- Liquid Engine: Applied DFM to manufacture 9 custom injector and combustion chamber components; used GD&T to ensure strict precision and tolerances for CNC/mill/lathe for successful static hot fire.

Mechanical Design Lead • Fremont High Robotics FRC Team 3501: Firebots

2020 - 2024

- Led a 6-person CAD team using Onshape, iterating a competition robot that placed in top 15% of 3,000+ teams worldwide.
- Managed all parts procurement (BOM), technical drawing approval, training program, and manufacturing workflow.

Personal Projects

ENGIN 100 Automated Rice Cooking System

- Automated standard rice cookers with IOT, coordinating 4 servos, pump, and a stirring motor via dual ESP32s over ESP-NOW; integrated load cell, ultrasonic, and optical sensors for automated rice/water level monitoring

ENGIN 29 — Collapsible Bike Trailer (Best Design & Integration Award)

- Designed and manufactured a collapsible bike trailer (¼ original footprint when stowed) with a tension-cable-triangulated aluminum/steel frame

ME130 — Automated Bow Maker (solo)

- Analytically synthesized a crank-rocker four-bar linkage and bevel gear transmission in MATLAB to satisfy a zero-point end-effector constraint and quick-return timing; optimized mechanical advantage to peak at the pinning stroke

Combat Robotics

- Designed, built, and tested multiple fully 3D-printed 1 lb combat robots for local competitions, modeling an offset slotted-link shuffler drivetrain mechanism in MATLAB and prototyping the crank-and-slider-based design.

Skills

CAE/Analysis: ANSYS, MATLAB, Python | **Design:** Onshape, SolidWorks, GD&T, DFM | **Shop:** TIG, CNC, Mill, Lathe, Carbon Fiber